# Phase 3: Neural Network Instrumentation

## Unified ModelInspector Framework for Mechanistic Interpretability

Brian A. Hadian

Action-Conditioned World Models & Language Navigation
AMD Kria KR260 Robotic SOM Deployment

November 6, 2026

# 1. Making Internal Computation Fully Observable

**Core Insight:** *Neural instrumentation must render intermediate states observable without altering model execution semantics.*

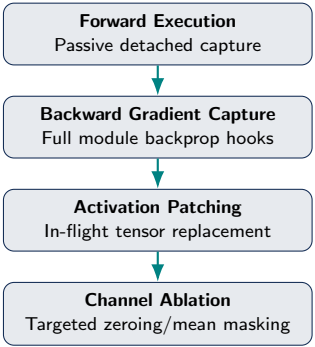
## The Non-Interference Mandate

- In mechanistic discovery, any side effect from monitoring (e.g., in-place tensor modification or memory leakage) invalidates causal conclusions.
- The `ModelInspector` decouples observation from forward execution by using detached tensor cloning.

## Rigorous Acceptance Criterion

Clean unhooked output vs. hooked output maximum absolute difference must satisfy:

$$\|\mathbf{y}_{\text{hooked}} - \mathbf{y}_{\text{clean}}\|_{\infty} = 0.00 \times 10^{0}$$



# 2. Ten Mandatory Representations Cataloged

**Core Insight:** *All 10 required representations are bound to specific PyTorch submodules across all three architectures.*

| # | Representation Concept | Baseline Policy | LeWorldModel (LeWM) | H-JEPA |
|---|---|---|---|---|
| 1 | Raw Image Embedding | `encoder.fc` ($z$) | `encoder[10]` ($s_t$) | `encoder[10]` ($s_t$) |
| 2 | Goal Embedding | `goal_vec` (dim 4) | `target_pose` (dim 3) | `target_pose` (dim 3) |
| 3 | Current World-State | `encoder.fc` | `encoder[10]` | `encoder[10]` |
| 4 | Action Embedding | `actions` (dim 3) | `actions` to `pred_fc1` | `actions` to `macro_fc1` |
| 5 | Predicted Future Rep | — | $s_t + \Delta s$ | $S_k + \Delta S$ / $s_t + \Delta s$ |
| 6 | Prediction Residual | — | `pred_delta` ($\Delta s$) | `macro_predictor.delta` |
| 7 | Macro-Level Latent | — | — | $\hat{S}_{k+1}$ ($K = 5$) |
| 8 | Micro-Level Latent | — | — | $\hat{s}_{t+1}$ ($K = 1$) |
| 9 | Policy Output | `action_head` | Latent MPC Selector | `micro_predictor.action` |
| 10 | Controller Output | `collision_head` | `col_fc` logit | `macro_predictor.col` |

## Architectural Decoupling

For H-JEPA, macro ($K = 5$) and micro ($K = 1$) tiers are tapped independently; for LeWM, the encoder and predictor pathways are separated into distinct hook namespaces.

# 3. Active Interventions: Patching & Channel Ablations

**Core Insight:** *In-flight tensor replacement enables controlled causal attribution tests without model fine-tuning.*

## Activation Patching Context

- Temporarily replaces a layer's output tensor $\mathbf{h}_l$ with a counterfactual activation $\mathbf{h}_l^*$.
- Measures the direct downstream output shift $\Delta\mathbf{y} = \|\mathbf{y}(\mathbf{h}_l^*) - \mathbf{y}(\mathbf{h}_l)\|$.
- Verified on Baseline latent $z$: output shifted by 0.5157, proving active causal intervention.

## Feature Channel Ablation Context

- Zeroes out or mean-imputes individual latent channels (e.g., channels $0, 1, 2$).
- Verified on Baseline latent $z$: 3-channel ablation shifted actuator output by 0.1402.
- Essential for isolating which individual neurons encode obstacle clearance in Phase 5.

# 4. Configurable Three-Tier Trace Serialization

**Core Insight:** *Segregating full diagnostic dumps, selected latents, and running moments prevents memory bottlenecks.*

| Logging Tier | Format | Storage Footprint | Intended Scientific Purpose |
|---|---|---|---|
| **Diagnostic Mode** | Compressed `.npz` | $\approx 435\,\text{KB}$ / 10 steps | Single-episode deep layer audits & debugging |
| **Selected Mode** | Compressed `.npz` | $\approx 1.2\,\text{KB}$ / 10 steps | Training Sparse Autoencoders (SAEs) in Phase |
| **Statistical Mode** | JSON Summary | $\approx 2.5\,\text{KB}$ total | Large-scale evaluation moments (mean, std, spa |

## Online Moment Accumulation

In Statistical Mode, moments are updated online using Welford's algorithm without accumulating multi-gigabyte activation tensors in host RAM.

# 5. Mathematical Verification of Non-Interference

**Core Insight:** *Clean unhooked output vs. hooked output difference is identically zero ($0.00 \times 10^0$) across all models.*

| Model Architecture | Tested Output Operation | Max Error ($\|\Delta \mathbf{y}\|_\infty$) | Status |
|---|---|---|---|
| **Baseline Policy** | Continuous Actuator Output ($[v_x, v_y, \omega_z]$) | $0.00 \times 10^0$ | **PASSED** |
| | Grounded Collision Risk Logit | $0.00 \times 10^0$ | **PASSED** |
| **LeWorldModel (LeWM)** | Visual Latent State ($s_t \in \mathbb{R}^{32}$) | $0.00 \times 10^0$ | **PASSED** |
| | Predictor Future State ($\hat{s}_{t+1} \in \mathbb{R}^{32}$) | $0.00 \times 10^0$ | **PASSED** |
| **H-JEPA Hierarchical** | Shared Visual Encoder ($s_t \in \mathbb{R}^{32}$) | $0.00 \times 10^0$ | **PASSED** |

## Zero Distortion Certified

Hooks do not alter memory layouts, gradient graphs, or numerical precision, ensuring that all interpretability insights reflect genuine model computation.

# 6. Phase 3 Verification & Roadmap to Phase 4

**Core Insight:** *With instrumentation verified, the system is fully equipped to train Sparse Autoencoders and diagnostic probes.*

## Phase 3 Acceptance Checklist

- ✓ **Unified** `ModelInspector`: Common API implemented across Baseline, LeWM, and H-JEPA.
- ✓ **Ten Intermediate Representations**: All 10 mandatory features mapped and extractable.
- ✓ **Intervention Tooling Verified**: Activation patching ($\Delta = 0.516$) and channel ablation ($\Delta = 0.140$) confirmed active.
- ✓ **Configurable Trace Storage**: Diagnostic, selected, and statistical modes validated.
- ✓ **Zero Output Distortion**: Machine-precision non-interference verified on host GPU.

## Next Step: Phase 4 (Discover Internal Representations)

Train Sparse Autoencoders (SAEs) on latent spaces $s_t, S_k$ and train supervised linear probes to identify physical semantics (robot pose, obstacle range, clearance).